

Database Management Systems: Assignment-12

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Q1 Perform the following tasks: (Use employee table from assignment 2)

1. Write a program in PL/SQL to display a cursor based detail information of employees from employees table.

Code:

```
DECLARE
    emp_cur IS
        SELECT emp_id, first_name, last_name, job_id, salary, dept_id
        FROM employee;

    v_emp_id    employee.emp_id%TYPE;
    v_fname     employee.first_name%TYPE;
    v_lname     employee.last_name%TYPE;
    v_job       employee.job_id%TYPE;
    v_salary    employee.salary%TYPE;
    v_dept      employee.dept_id%TYPE;
BEGIN
    OPEN emp_cur;

    LOOP
        FETCH emp_cur INTO v_emp_id, v_fname, v_lname, v_job, v_salary, v_dept;
        EXIT WHEN emp_cur%NOTFOUND;

        DBMS_OUTPUT.PUT_LINE(
            'ID: ' || v_emp_id ||
            ', Name: ' || v_fname || ' ' || v_lname ||
            ', Job: ' || v_job ||
            ', Salary: ' || v_salary ||
            ', Dept: ' || v_dept
        );
    END LOOP;

    CLOSE emp_cur;
END;
```

/

Output:

```
ID: 103, Name: Bob Johnson, Job: SA_REP, Salary: 52000, Dept: 30
ID: 101, Name: John Doe, Job: IT_PROG, Salary: 60000, Dept: 10
ID: 105, Name: Aman Patel, Job: FIN, Salary: 48000, Dept: 40
ID: 102, Name: Alice Smith, Job: HR, Salary: 45000, Dept: 20
ID: 104, Name: Riya Sharma, Job: IT_PROG, Salary: 70000, Dept: 10
```

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.035

2. Write a program in PL/SQL to retrieve the records from the employees table and display them using cursors.

Code:

```
DECLARE
    CURSOR c_emp IS
        SELECT * FROM employee;

    v_emp employee%ROWTYPE;
BEGIN
    OPEN c_emp;

    LOOP
        FETCH c_emp INTO v_emp;
        EXIT WHEN c_emp%NOTFOUND;

        DBMS_OUTPUT.PUT_LINE(
            v_emp.emp_id || ' - ' ||
            v_emp.first_name || ' ' || v_emp.last_name ||
            ' - Job: ' || v_emp.job_id ||
            ' - Salary: ' || v_emp.salary ||
            ' - Dept: ' || v_emp.dept_id
        );
    END LOOP;

    CLOSE c_emp;
END;
```

/

Output:

```
103 - Bob Johnson - Job: SA_REP - Salary: 52000 - Dept: 30
101 - John Doe - Job: IT_PROG - Salary: 60000 - Dept: 10
105 - Aman Patel - Job: FIN - Salary: 48000 - Dept: 40
102 - Alice Smith - Job: HR - Salary: 45000 - Dept: 20
104 - Riya Sharma - Job: IT_PROG - Salary: 70000 - Dept: 10
```

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.007

3. Write a program in PL/SQL to create an implicit cursor with for loop.

Code:

```
BEGIN
  FOR r IN (SELECT emp_id, first_name, last_name, salary FROM employee)
  LOOP
    DBMS_OUTPUT.PUT_LINE(
      r.emp_id || ' - ' || r.first_name || ' ' ||
      r.last_name || ' - Salary: ' || r.salary
    );
  END LOOP;
END;
/
```

Output:

```
103 - Bob Johnson - Salary: 52000
101 - John Doe - Salary: 60000
105 - Aman Patel - Salary: 48000
102 - Alice Smith - Salary: 45000
104 - Riya Sharma - Salary: 70000
```

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.010

4. Write a program in PL/SQL to create an explicit cursor with for loop.

Code:

```
DECLARE
  CURSOR emp_cursor IS
    SELECT emp_id, first_name, last_name, job_id
    FROM employee;
BEGIN
  FOR rec IN emp_cursor LOOP
    DBMS_OUTPUT.PUT_LINE(
      rec.emp_id || ' - ' || rec.first_name || ' ' ||
      rec.last_name || ' - Job: ' || rec.job_id
    );
  END LOOP;
END;
/
```

Output:

```
103 - Bob Johnson - Job: SA_REP
101 - John Doe - Job: IT_PROG
105 - Aman Patel - Job: FIN
102 - Alice Smith - Job: HR
104 - Riya Sharma - Job: IT_PROG
```

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.040

Q2 Considering the tables of Assignment 9 (Question-3), perform the following tasks:

1. Create a trigger to update the remaining quantity of books in the ORDER_DETAILS table, when a new entry is inserted.

Code:

```
CREATE OR REPLACE TRIGGER trg_update_remaining_qty
AFTER INSERT ON order_details
FOR EACH ROW
BEGIN
    UPDATE books
    SET qty_available = qty_available - :NEW.qty_ordered
    WHERE book_id = :NEW.book_id;
END;
/
```

Output:

```
SQL> CREATE OR REPLACE TRIGGER trg_update_remaining_qty
      AFTER INSERT ON order_details
      FOR EACH ROW
      BEGIN...
Show more...
```

Trigger TRG_UPDATE_REMAINING_QTY compiled

No errors.

Elapsed: 00:00:00.018

2. Create a trigger to find the difference between old no. of quantities of books and new no. of quantities of books.

Code:

```
CREATE OR REPLACE TRIGGER trg_qty_difference
BEFORE UPDATE ON order_details
FOR EACH ROW
DECLARE
    diff NUMBER;
BEGIN
    diff := :NEW.qty_ordered - :OLD.qty_ordered;

    DBMS_OUTPUT.PUT_LINE('Quantity difference: ' || diff);

    UPDATE books
    SET qty_available = qty_available - diff
    WHERE book_id = :NEW.book_id;
END;
/
```

Output:

```
SQL> CREATE OR REPLACE TRIGGER trg_qty_difference
      BEFORE UPDATE ON order_details
      FOR EACH ROW
      DECLARE...
      Show more...
```

Trigger TRG_QTY_DIFFERENCE compiled

No errors.

Elapsed: 00:00:00.010